Logic Minimization – Part 3

K. Sridharan

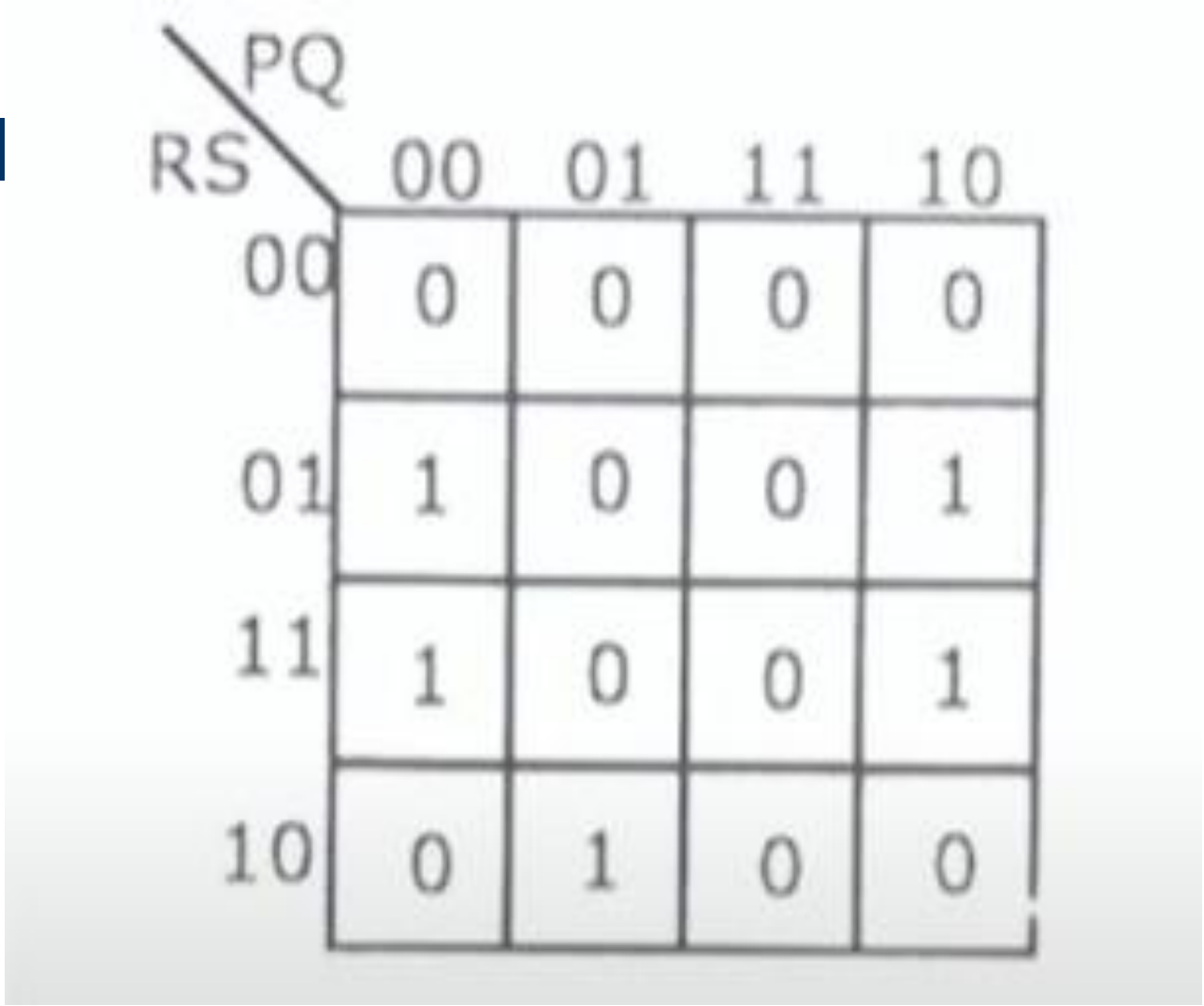
Review

- We have discussed K-maps for simplification.
- We will look at one more scenario with K-maps involving POS form.
- Then we will discuss a notion called implicant and variations called prime implicant and essential prime implicant.

Can we obtain the minimal POS form using a K-map ?

- Recall that we have discussed about product of sum (POS) form earlier in the context of expressing Boolean functions as the product of maxterms.
- The question then is: Can we do simplification using a K-map and get a simplified POS form for a Boolean function ?
- Indeed, we can. We will illustrate this possibility via an example.

Example ..



RS \ PQ	00	01	11	10
00	0	0	0	0
01	1	0	0	1
11	1	0	0	1
10	0	1	0	0

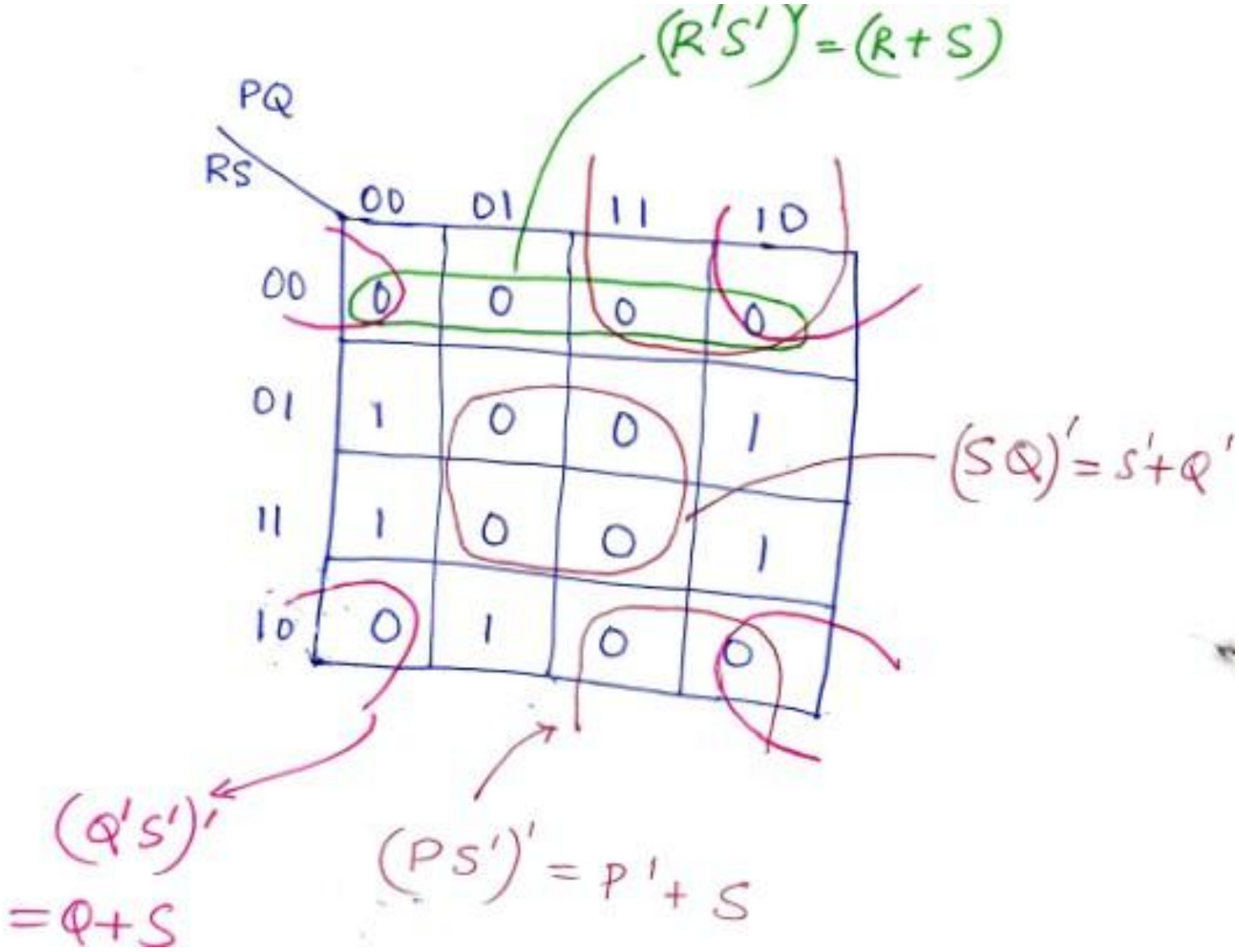
Question for students

- We are given a plot of the minterms. How do we go about extracting the POS form ?

Use the 0's in the plot

- Identify the cells that have an entry of 0 and simplify as before.
- However, we need to carefully write each of the resulting terms. The solution is shown next.

The K-map


(R'S') = (R+S)
PQ
RS
00
01
11
10
00
01
11
10
0 0 0 0
1 0 0 1
1 0 0 1
0 1 0 0
(SQ)' = S'+Q'
(Q'S')'
=Q+S
(PS')' = P'+S

The function is therefore ..

- F = (R + S) (S’ + Q’) (S + P’)(S + Q)

- We now move on to discuss about implicants and variations. We will also show how to use a K-map to identify these.

Implicants

- Given a function F of n variables, a product term P is an implicant of F if and only if for every combination of values of the n variables for which P =1, F is also equal to 1.
- That is, if for some combination of values of the variables, P = 1 and F=0, then P is not an implicant of F. Consider F = a'b'c' + ab'c' + ab'c + abc. Here a'b'c' is an implicant while bc is not an implicant. When a'b'c' =1, note F = 1. However, if a = 0, b=1 and c= 1, then bc = 1 but note that F will not be 1.

Exercise

- Show that *ac* is an implicant of F.

  Hint: All we need to see is if *ac* =1 implies F=1 (allowing *b* to take a value of 0 or 1)

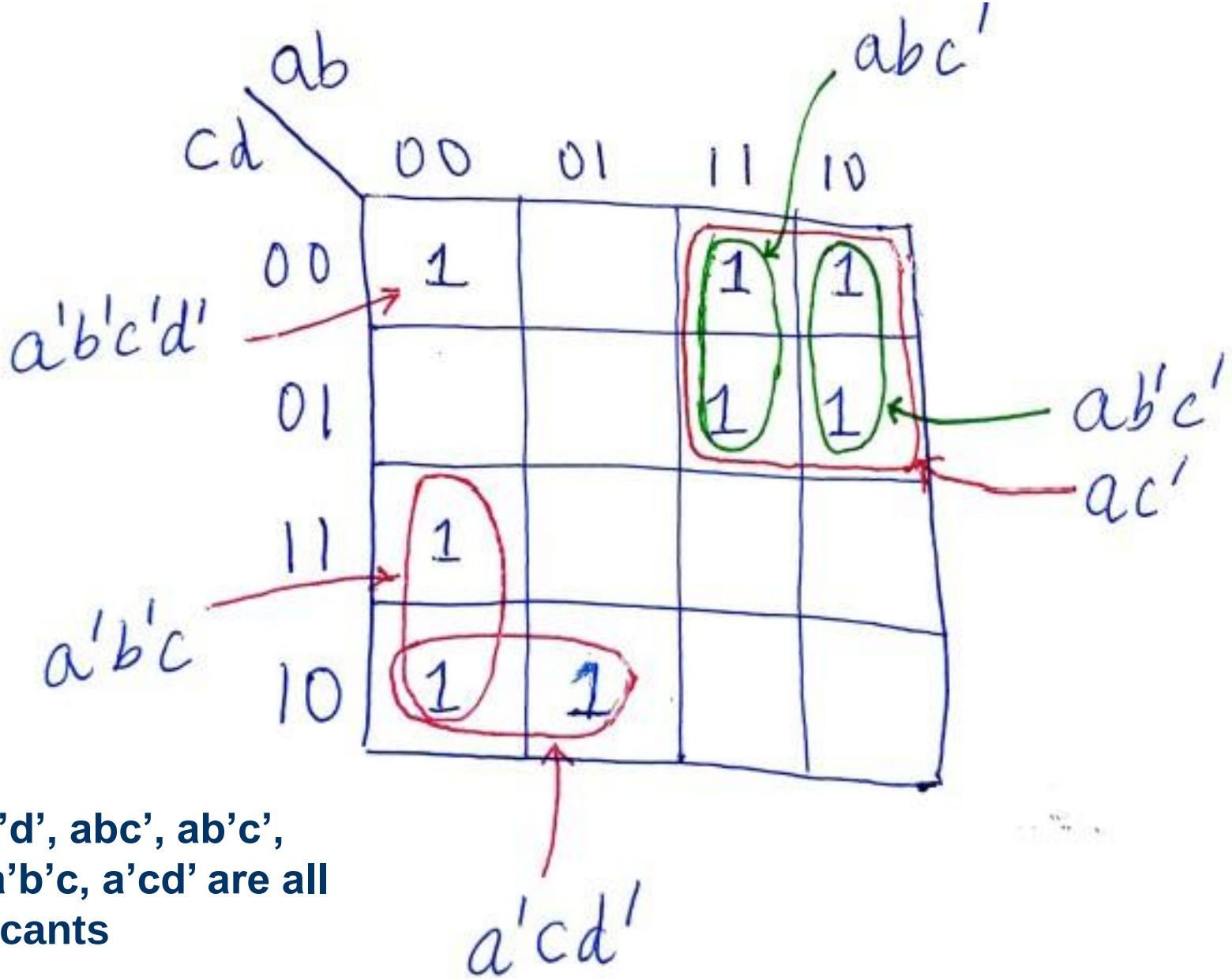
Why did we choose ac ?

- Note that F(a,b,c) is also equal to b’c’ + ac
  (verify that a’b’c’+ ab’c’ + ab’c + abc leads to this)

More on Implicants

- In sum-of-products form, every product term is an implicant. Every minterm of a function F is also an implicant. So is any term formed by combining two or more minterms.

What is an Implicant in a K-map ?



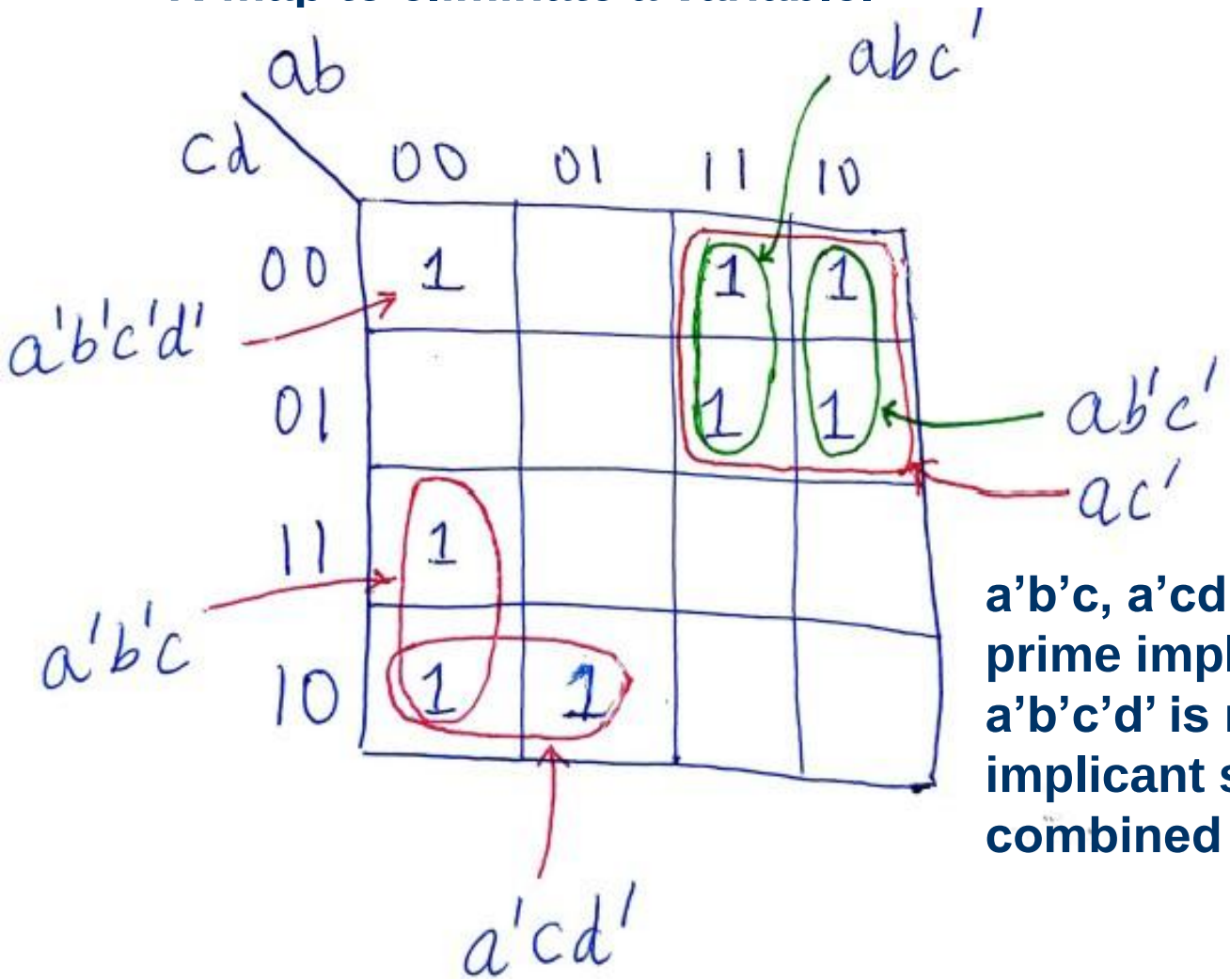
a'b'c'd', abc', ab'c', ac', a'b'c, a'cd' are all implicants

Prime Implicants

- A prime implicant of a function F is a product term implicant which is no longer an implicant if any literal (a variable or its complement) is deleted from it.
- In the function F = a'b'c' + ab'c' + ab'c + abc, implicant a'b'c' is not a prime implicant since a' can be removed (and the resulting term b'c' is still an implicant of F).
- However, b'c' is a prime implicant since deletion of b' or c' will not allow the remaining to be an implicant. That is if b' is removed, c' alone is not an implicant (similarly, b' alone is not an implicant). Note that F = b'c' + ac

What is a Prime Implicant in a K-map ?

A prime implicant is a product term implicant if it cannot be combined with any other term in the K-map to eliminate a variable.



a’b’c, a’cd’ and ac’ are prime implicants but a’b’c’d’ is not a prime implicant since it can be combined with a’b’cd’

Question for students

- Is abc’ is a prime implicant ?
- What about ab’c’ ?

None of the two are prime implicants !

- abc’ is not a prime implicant
- Similarly, ab’c’ is not a prime implicant.
- The reason is both of these can combine together to form ac’ (see the K-map shown earlier)

Notes

- Note 1: the 1 in the top cell (corresponding to 0000) and the 1 in the bottom cell (corresponding to 0010) in the same column are adjacent.
- Note 2: Think of a prime implicant as a product term obtained by combining the max possible number of adjacent squares in the K-Map.

Remarks on prime implicants

- All of the prime implicants of a function can be obtained from a K-map.
- A single 1 represents a prime implicant if it is not adjacent to any other 1's.
- Two adjacent 1's form a prime implicant if they are not contained in a group of four 1's.
- Four adjacent 1's form a prime implicant if they are not contained in a group of eight 1's.

What happens if don’t cares exist ?

- In the process of finding prime implicants, don’t cares are treated just like 1’s.
- However, a prime implicant composed entirely of don’t cares can never be part of the minimum solution.

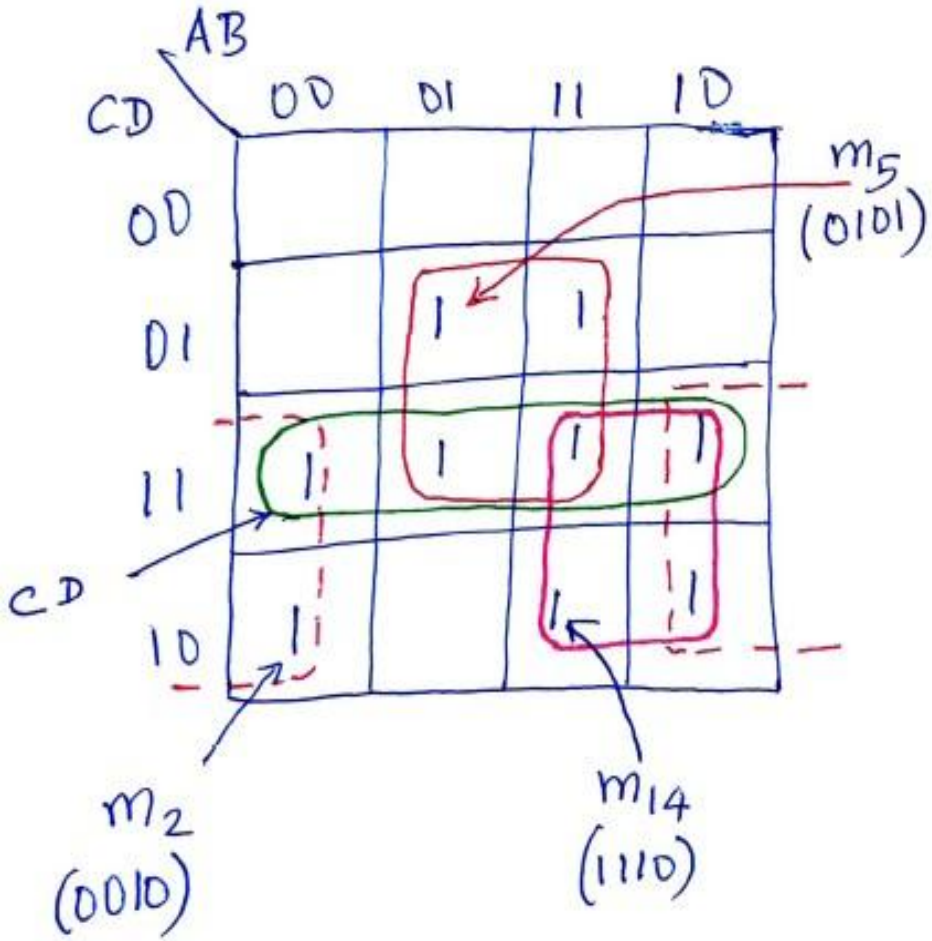
Are all prime implicants needed in simplification ?

- No, some of the prime implicants of a function may not be required for forming the minimum sum of products.
- We will now discuss the notion of essential prime implicants.

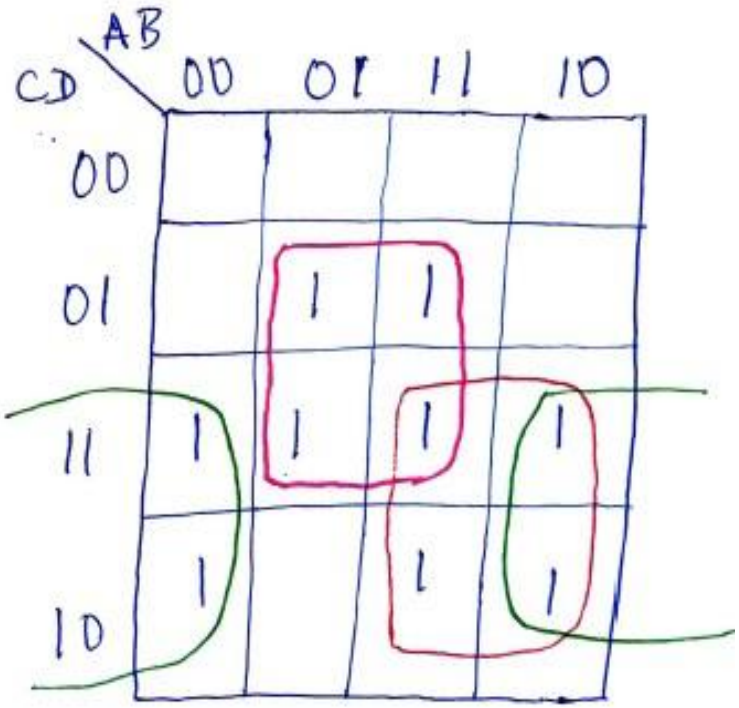
Essential Prime Implicants

- In general, some of the minterms in a K-map can be covered by only a single prime implicant while other minterms may be covered by more than one prime implicant.
- If a minterm is covered by only one prime implicant, then that prime implicant is termed as essential prime implicant.
- We illustrate this next.

Two possibilities for simplification



$f = CD + BD + B'C + AC$



$f = BD + B'C + AC$

LHS leads to wrong min sum of products since CD is not required.

How do we discover this?

- If we choose CD first, then BD, B'C and AC are needed to cover the remaining 1's so we will have four terms in the solution.
- Note that m_2 (corresponding to 0010) is covered only by B'C so B'C is an essential prime implicant. Also, only prime implicant which covers m_5 is BD so BD is essential. Also, AC is essential since m_{14} is covered by only AC.
- However, CD is not essential since each of the 1's in CD is covered by another prime implicant.

How else could CD be neglected ?

- Question for students

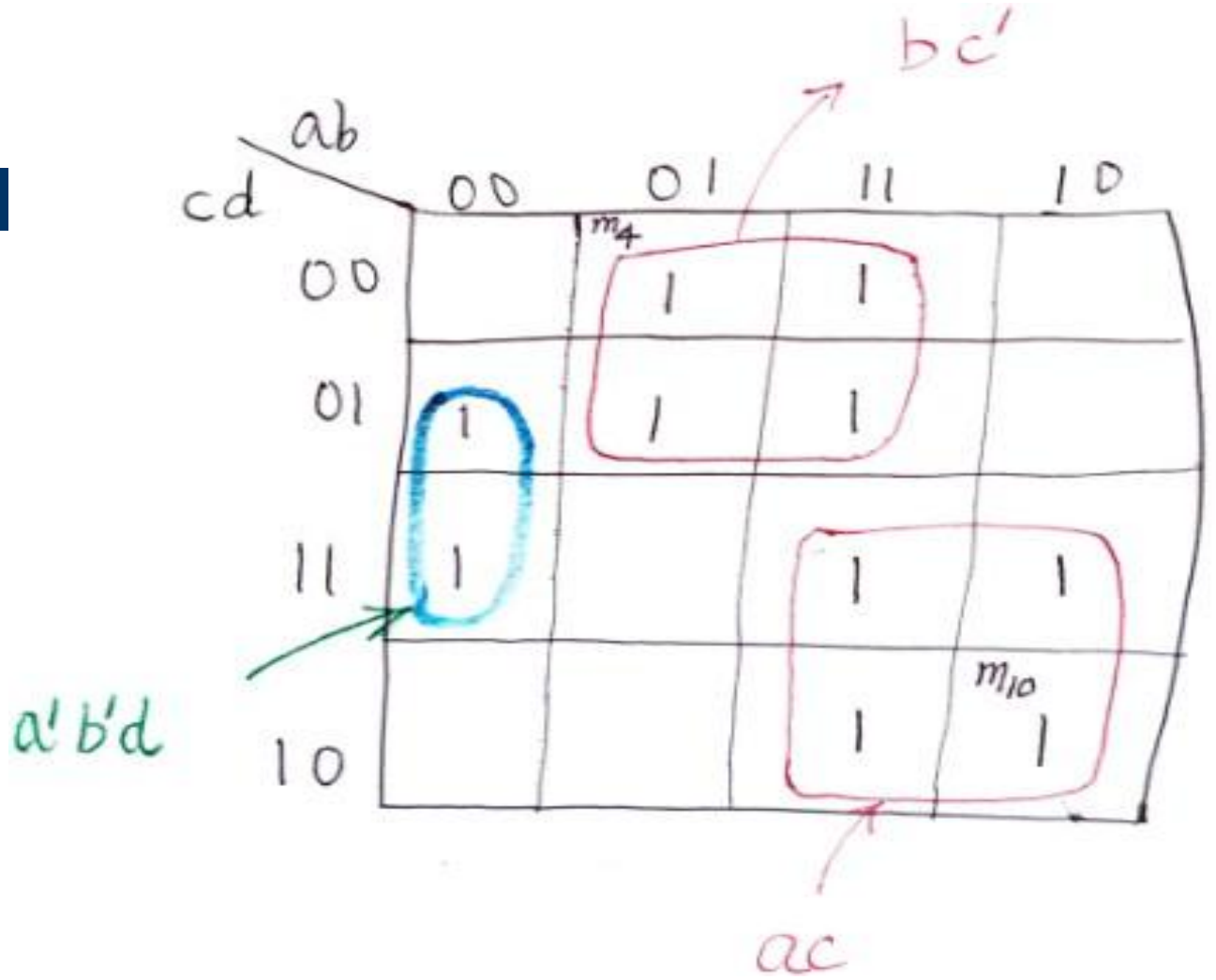
How can we discard CD ?

- Note that CD is the consensus term (see also Tutorial 2 for a question on this).
- This, of course, becomes clear only if we had both the expressions (the one with CD and the one without CD).

Will the final solution have only essential prime implicants ?

- Not necessary.
- In general, the overall minimum solution can have essential prime implicants and some other terms (prime implicants).
- The prime implicant list itself can be quite large.
- We illustrate these ideas via an example next.

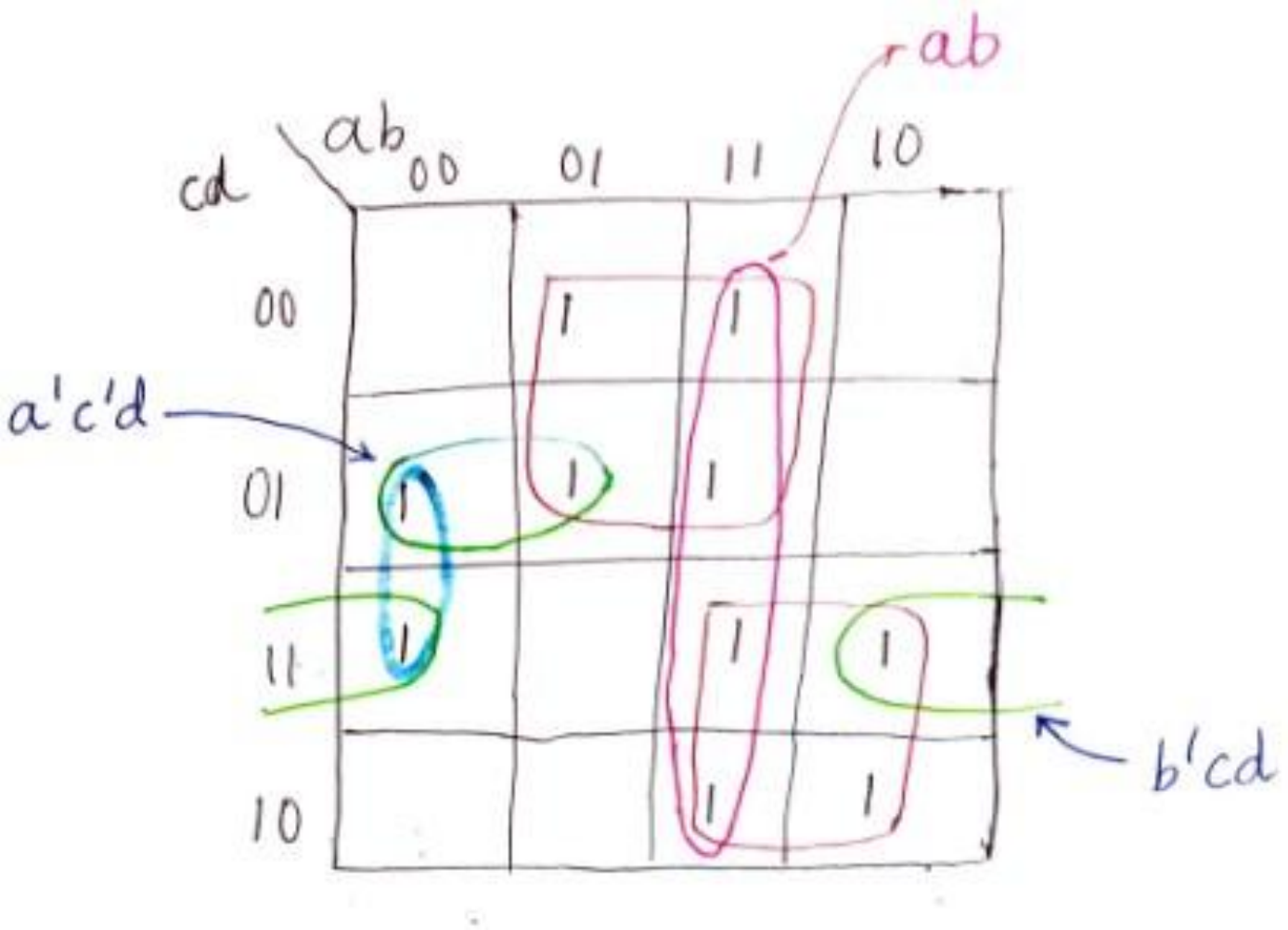
K-map and minimum solution



Explanation

- A plot of the minterms and the minimum solution is shown: F = a'b'd + bc' + ac
- Of these, bc' and ac are the only essential prime implicants since m_4 is covered only by bc' while m_{10} is covered by only ac
- The third term, namely a'b'd has 1's which are also part of other prime implicants such as a'c'd and b'cd as shown next.

Other prime implicants marked



Remarks

- This example shows that the list of prime implicants can be much larger than the list of essential prime implicants.
- Further, only some of the prime implicants are part of the (final) simplified solution.
- Also, only some of the terms (prime implicants) in the final solution are essential prime implicants.

Textbook and Reference

- M. Mano and M. Ciletti, Digital Design, Pearson Education, 2018
- C. Roth and L.L. Kinney, Fundamentals of Logic Design, Gamma Publications, 2021